Description

Illustration



Figure caption

1. Sandstone bricks. 2. Traditional Maltese building made of sandstone. 3. Weathered sandstone on a façade. Images used under license from Shutterstock.com

The material

Sandstone is consolidated sand particles (quartz), bonded by a cementing agent: feldspars, limes, silica or clays. The size of the sand particles, the porosity and the strength vary greatly in different sandstones. The colors derive from iron or manganese impurities and give sandstones their character.

Compositional summary

Silica (SiO2) particles bonded with lime (CaO), calcium carbonate (CaCO3) or clays (alumino-silicates).

General properties

Density		2.24e3	-	2.65e3	kg/m^3
Price	*	0.31	-	0.468	GBP/kg
Date first used		-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✓

Mechanical properties

Young's modulus		14	-	25	GPa

Shear modulus		5.6	-	10	GPa
Bulk modulus		11	-	20	GPa
Bending modulus	*	14	-	40	GPa
Poisson's ratio		0.22	-	0.29	
Yield strength (elastic limit)		4	-	22	MPa
Tensile strength		4	-	22	MPa
Compressive strength	*	50	-	155	MPa
Bending strength		5	-	16	MPa
Elongation	*	0.02	-	0.16	% strain
Hardness - Vickers		7	-	38	HV
Fatigue strength at 10^7 cycles	*	3.1	-	12	MPa
Fracture toughness	*	0.7	-	1.1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0019	-	0.0057	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity	*	0.2	-	1.11	m.°C/W
Thermal expansion coefficient	*	8	-	20	µstrain/°C
Specific heat capacity	*	840	-	920	J/kg.°C
Melting point	*	1.2e3	-	1.4e3	°C
Maximum service temperature	*	400	-	600	°C
Minimum service temperature		-273			°C
Flammability		Non-flammable			
Emissivity		0.6	-	0.83	

Hygro-thermal properties

Water absorption		2	-	8.5	%
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1e10	-	1e14	µohm.cm
Dielectric constant (relative permittivity)	*	6	-	9	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)		5	-	12	MV/m

Optical properties

Natural color

Pure siliceous sandstone is white or cream. But red, gray, blueish or black sandstones are common.

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Machinability	3	-	4	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Acceptable
Strong acids	Acceptable
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Limited use
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Acceptable

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	*	0.0571	-	0.0629	kg/kg
Embodied energy (virgin grade production)	*	0.951	-	1.05	MJ/kg
Water Usage	*	3.23	-	3.57	l/kg

Material processing: energy

Machining, grinding, embodied energy	*	21.1	-	34.9	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO_2-eq)	*	1.38	-	2.29	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	1.34	-	1.48	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Sandstone is easily cut and carved. It weathers in a benign attractive way, but the surface traps dirt in an urban or industrial environment, requiring periodic cleaning.

Technical notes

Sandstones consist of particles of quartz, feldspar and mica bonded by a natural cement. The cement determines the strength, durability and color. Calcareous sandstones are bonded with calcium carbonate; they are called "freestone" because they are easily worked but they weather badly. Siliceous sandstones are bonded with alumino-silicates; they are acid resistant and durable but harder to work. Bluestone, much used in New York state, is noted for its even grain and high strength. It is about 70% silica bonded with clay. Ferruginous sandstones contain oxides of iron, giving lovely browns, reds and yellows.

Typical uses

Buildings and facing, table tops, bench tops and chemical equipment to resist acids and alkalis.

Tradenames

York stone; Bluestone

Related standards

04400 Stone
04410 Rough Stone
04420 Cut Stone
04440 Flagstone
04450 Stone Veneer
04470 Sandstone

Links

ProcessUniverse

Reference